2023 Ch H2 Q2

Section: Chemical Changes and Structure

Topic: Structure and Bonding / Energetics

Question Summary

(a) Hydrides of group 4 and 5:

(i) Explain the increase in boiling points of group 4 hydrides down the group (refer to intermolecular forces).

(ii) Name the intermolecular force responsible for the anomalous boiling point of NH_3.

(b) Silicon hydride, SiH_4:

(i) Use the given thermochemical data to calculate ΔH for $Si(s) + 2H_2(g) \rightarrow SiH_4(g)$.

(ii) From 15.32 g Mg_2Si (excess HCl), 2.56 g SiH_4 is produced. Calculate percentage yield.

(iii) Explain fully why SiO_2 has a much higher melting point than SiH_4.

Worked Solution

According to the SQA Marking Instructions for 2023 H2 Q2:

(a)(i) Boiling points of the group 4 hydrides (CH_4 → SiH_4 → GeH_4 → SnH_4) increase down the group because London dispersion forces become stronger as molar mass and polarisability increase. Larger electron clouds are more easily distorted, leading to greater instantaneous dipoles and stronger attractions between molecules.

(a)(ii) The anomalously high boiling point of ammonia is due to hydrogen bonding between NH_3 molecules (H attached to highly electronegative N with lone pairs).

(b)(i) Use Hess' law. Given:
• $SiH_4 + 2O_2 \rightarrow SiO_2 + 2H_2O$, $\Delta H = -1517$ kJ mol^{-1}
• $Si + O_2 \rightarrow SiO_2$, $\Delta H = -911$ kJ mol^{-1}
• $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$, $\Delta H = -286$ kJ mol^{-1} (so $2H_2 \rightarrow 2H_2O$ is -572 kJ mol^{-1})
Combine ($Si + 2H_2 + 2O_2 \rightarrow SiO_2 + 2H_2O$) with $\Delta H = -1483$, then subtract the combustion of SiH_4 (-1517).

Result for $Si + 2H_2 \rightarrow SiH_4$: $\Delta H = +34$ kJ mol⁻¹ (endothermic).

(b)(ii) Moles of Mg_2Si = 15.32 g ÷ 76.7 g mol^{-1} = 0.200 mol (≈ 0.1997). Stoichiometry $Mg_2Si : SiH_4$ is 1 : 1 → theoretical moles SiH_4 = 0.200 mol. Theoretical mass = 0.200 × 32.1 = 6.42 g (≈ 6.40 g). Actual mass = 2.56 g. Percentage yield = 2.56 ÷ 6.40 × 100 = 40.0%.

(b)(iii) SiO_2 is a covalent network solid with a giant lattice of strong Si–O covalent bonds throughout the structure; a very large amount of energy is needed to break this network → very high melting point (1710 °C). SiH_4 is a simple covalent molecular substance; only weak London dispersion forces exist between molecules, so it melts at a very low temperature (−185 °C).

Final Answer

(a)(i) Boiling points increase down group 4 due to stronger London dispersion forces (greater size/polarisability).

(a)(ii) Hydrogen bonding.

(b)(i) ΔH($Si + 2H_2 \rightarrow SiH_4$) = +34 kJ mol^{-1}.

(b)(ii) Percentage yield = 40.0%.

(b)(iii) SiO_2: covalent network (many strong Si–O bonds) ≫ SiH_4: simple molecular (weak LDF) → much higher m.p. for SiO_2.

Revision Tips

- Down a group: polarisability ↑ → LDF ↑ → b.p. ↑ (for similar molecules).
- Hydrogen bonding needs H–N/O/F and lone pairs.
- Use Hess' cycles systematically; align equations and cancel species.
- Yield = (actual ÷ theoretical) × 100.